

References

- [1] Achdou, Y. and Pironneau, O. (2005). *Computational methods for option pricing*, volume 30 of *Frontiers in Applied Mathematics*. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA. (Cited on page 377).
- [2] Ackerer, D., Tagasovska, N., and Vatter, T. (2020). Deep smoothing of the implied volatility surface. In *Proceedings of the 34th Conference on Neural Information Processing Systems (NeurIPS)*, pages 11552–11563. (Cited on page 377).
- [3] Albanese, C. and Lawi, S. (2005). Laplace transforms for integrals of Markov processes. *Markov Process. Related Fields*, 11(4):677–724. (Cited on pages 330 and 614).
- [4] Albrecher, H., Mayer, P., Schoutens, W., and Tistaert, J. (2007). The little Heston trap. *Wilmott Magazine*, pages 83–92. (Cited on page 348).
- [5] Allegretto, W., Barone-Adesi, G., and Elliott, R. (1995). Numerical evaluation of the critical price and American options. *European Journal of Finance*, 1:69–78. (Cited on page 559).
- [6] Applebaum, D. (2009). *Lévy processes and stochastic calculus*, volume 116 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, second edition. (Cited on page 760).
- [7] Aristotle. *Politics, Book one, Part XI*. The Internet Classics Archive, 350 BCE. <http://classics.mit.edu/Aristotle/politics.1.one.html>. (Cited on page 5).
- [8] Armstrong, J. and Tatlow, G. (2024). Deep gamma hedging. Preprint arXiv:2409.13567. (Cited on page 272).
- [9] Attari, M. (2004). Option pricing using Fourier transforms: a numerically efficient simplification. Preprint, 7 pages. (Cited on page 348).
- [10] Bachelier, L. (1900). Théorie de la spéculation. *Annales Scientifiques de l'Ecole Normale Supérieure, Série 3*, 17:21–86. (Cited on pages 2, 3, 158, 216, 263, 270, 299, and 306).

- [11] Barone-Adesi, G. and Whaley, R. (1987). Efficient analytic approximation of American option values. *The Journal of Finance*, 42(2):301–320. (Cited on pages 559, 561, and 1143).
- [12] Barrieu, P., Rouault, A., and Yor, M. (2004). A study of the Hartman-Watson distribution motivated by numerical problems related to the pricing of Asian options. *J. Appl. Probab.*, 41(4):1049–1058. (Cited on page 488).
- [13] Bartlett, B. (2006). Hedging under SABR model. *Wilmott Magazine*, pages 2–4. (Cited on page 1044).
- [14] Bergomi, L. (2016). *Stochastic Volatility Modeling*. Financial Mathematics Series. Chapman & Hall/CRC. (Cited on page 1051).
- [15] Bermin, H. (1998). Essays on lookback options: a Malliavin calculus approach. PhD thesis, Lund University. (Cited on page 469).
- [16] Bickersteth, M., Ding, Y., and Rutkowski, M. (2025). Pricing and hedging of SOFR derivatives. *Math. Finance*, to appear. (Cited on page 668).
- [17] Billingsley, R. and Chance, D. (1985). Options market efficiency and the box spread strategy. *The Review of Financial Studies*, 20(4):287–301. (Cited on page 995).
- [18] Björk, T. (2004). *Arbitrage Theory in Continuous Time*, volume 121 of *Oxford Finance*. Oxford University Press. (Cited on page 44).
- [19] Björk, T. and Christensen, B. (1999). Interest rate dynamics and consistent forward rate curves. *Math. Finance*, 9(4):323–348. (Cited on page 681).
- [20] Black, F. (1976). The pricing of commodity contracts. *J. of Financial Economics*, 3:167–179. (Cited on pages 704, 705, and 719).
- [21] Black, F., Derman, E., and Toy, W. (1990). A one-factor model of interest rates and its application to treasury bond options. *Financial Analysts Journal*, 46(1):24–32. (Cited on page 651).
- [22] Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *J. of Political Economy*, 81. (Cited on pages 3, 4, 210, 231, and 233).
- [23] Borodin, A. (2017). *Stochastic processes*. Probability and its Applications. Birkhäuser/Springer, Cham. Original Russian edition published by LAN Publishing, St. Petersburg, 2013. (Cited on page 401).
- [24] Bosq, D. and Nguyen, H. (1996). *A Course in Stochastic Processes: Stochastic Models and Statistical Inference*. Mathematical and Statistical Methods. Kluwer. (Cited on page 737).
- [25] Boulding, K. (1973). In “Energy Reorganization Act of 1973. Hearings, Ninety-third Congress, first session, on H.R. 11510”. U.S. Government Printing Office, Washington. (Cited on pages 213 and 574).
- [26] Boyle, P. and Vorst, T. (1992). Option replication in discrete time with transaction costs. *The Journal of Finance*, XLVII(1):271–293. (Cited on page 907).

- [27] Brace, A., Gatarek, D., and Musiela, M. (1997). The market model of interest rate dynamics. *Math. Finance*, 7(2):127–155. (Cited on pages 4 and 687).
- [28] Breeden, D. and Litzenberger, R. (1978). Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51:621–651. (Cited on page 373).
- [29] Brémaud, P. (1999). *Markov chains*, volume 31 of *Texts in Applied Mathematics*. Springer-Verlag, New York. (Cited on page 752).
- [30] Brigo, D. and Mercurio, F. (2006). *Interest rate models—theory and practice*. Springer Finance. Springer-Verlag, Berlin, second edition. (Cited on pages 341, 630, 686, and 1193).
- [31] Briola, A., Vidal-Tomás, D., Wang, Y., and Aste, T. (2023). Anatomy of a stablecoin’s failure: The Terra-Luna case. *Finance Research Letters*, 51:103358. (Cited on page 229).
- [32] Broadie, M. and Detemple, J. (1996). American option valuation: New bounds, approximations, and a comparison of existing methods. *The Review of Financial Studies*, 9:1211–1250. (Cited on page 565).
- [33] Broadie, M. and Jain, A. (2008). The effect of jumps and discrete sampling on volatility and variance swaps. *Int. J. Theor. Appl. Finance*, 11(8):761–797. (Cited on page 358).
- [34] Brody, D., Hughston, L., and Meier, D. (2018). Lévy-Vasicek models and the long-bond return process. *Int. J. Theor. Appl. Finance*, 21(3):1850026. (Cited on pages 645, 1186, and 1188).
- [35] Brown, C., Handley, J., Ling, C.-T., and Palmer, K. (2016). Partial differential equations for Asian option prices. *Quant. Finance*, 16(3):447–460. (Cited on page 511).
- [36] Brown, R. (1828). A brief account of microscopical observations made in the months of June, July and August, 1827, on the particles contained in the pollen of plants; and on the general existence of active molecules in organic and inorganic bodies. *Philosophical Magazine*, 4:161–173. (Cited on pages 1 and 152).
- [37] Buchen, P. (2012). *An introduction to exotic option pricing*. Chapman & Hall/CRC Financial Mathematics Series. CRC Press, Boca Raton, FL. (Cited on page 1106).
- [38] Burdzy, K. (1990). On nonincrease of Brownian motion. *Ann. Probab.*, 18(3):978–980. (Cited on page 390).
- [39] Carmona, R. and Durrleman, V. (2003). Pricing and hedging spread options. *SIAM Rev.*, 45(4):627–685. (Cited on pages 307, 1007, and 1211).
- [40] Carr, P. and Lee, R. (2008). Robust replication of volatility derivatives. Mathematics in Finance Working Paper Series, Working Paper #2008-3. (Cited on pages 334 and 359).
- [41] Carr, P. and Schröder, M. (2004). Bessel processes, the integral of geometric Brownian motion, and Asian options. *Theory Probab. Appl.*, 48(3):400–425. (Cited on pages 489 and 490).

- [42] Çinlar, E. (2011). *Probability and Stochastics*, volume 261 of *Graduate Texts in Mathematics*. Springer, New York. (Cited on page 829).
- [43] Chan, K., Karolyi, G., Longstaff, F., and Sanders, A. (1992). An empirical comparison of alternative models of the short-term interest rate. *The Journal of Finance*, 47(3):1209–1227. Papers and Proceedings of the Fifty-Second Annual Meeting of the American Finance Association, New Orleans, Louisiana. (Cited on pages 644 and 646).
- [44] Charlier, C. (1914). Frequency curves of type A in heterograde statistics. *Ark. Mat. Astr. Fysik*, 9(25):1–17. (Cited on page 790).
- [45] Charpentier, A., editor (2014). *Computational Actuarial Science with R. The R Series*. Chapman & Hall/CRC, USA. (Cited on pages 663 and 682).
- [46] Chataigner, M., Cousin, A., Crépey, S., Dixon, M., and Gueye, D. (2021). Beyond surrogate modeling: Learning the local volatility via shape constraints. *SIAM Journal on Financial Mathematics*, 12(3):SC58–SC69. (Cited on page 377).
- [47] Christensen, J., Diebold, F., and Rudebusch, G. (2011). The affine arbitrage-free class of Nelson-Siegel term structure models. *Journal of Econometrics*, 164:4–20. (Cited on page 682).
- [48] Clark, M. (2000). Trillion Dollar Bet. PBS Nova Documentaries. (Cited on page 3).
- [49] Cont, R. and Tankov, P. (2004). *Financial modelling with jump processes*. Chapman & Hall/CRC Financial Mathematics Series. Chapman & Hall/CRC, Boca Raton, FL. (Cited on pages 760, 768, 774, 783, and 805).
- [50] Courtadon, G. (1982). The pricing of options on default-free bonds. *The Journal of Financial and Quantitative Analysis*, 17(1):75–100. (Cited on pages 617 and 646).
- [51] Cox, A. and Hobson, D. (2005). Local martingales, bubbles and option prices. *Finance Stoch.*, 9(4):477–492. (Cited on page 315).
- [52] Cox, J., Ingersoll, J., and Ross, S. (1985). A theory of the term structure of interest rates. *Econometrica*, 53:385–407. (Cited on pages 264, 307, 330, 341, 614, 615, 618, 631, 641, and 647).
- [53] Cox, J., Ross, S., and Rubinstein, M. (1979). Option pricing: A simplified approach. *Journal of Financial Economics*, 7:87–106. (Cited on pages 74, 95, 99, and 996).
- [54] Cramér, H. (1946). *Mathematical methods of statistics*. Princeton University Press, Princeton, NJ. (Cited on page 790).
- [55] Crépey, S. (2013). *Financial modeling*. Springer Finance. Springer, Heidelberg. A backward stochastic differential equations perspective, Springer Finance Textbooks. (Cited on page 480).
- [56] Curran, M. (1994). Valuing Asian and portfolio options by conditioning on the geometric mean price. *Management Science*, 40(12):1705–1711. (Cited on page 495).
- [57] Dahl, L. and Benth, F. (2002). Fast evaluation of the Asian basket option by singular value decomposition. In *Monte Carlo and quasi-Monte Carlo*

- methods, 2000 (Hong Kong)*, pages 201–214. Springer, Berlin. (Cited on page 497).
- [58] Dana, R.-A. and Jeanblanc, M. (2007). *Financial markets in continuous time*. Springer Finance. Springer-Verlag, Berlin. Corrected Second Printing. (Cited on page 462).
- [59] Dash, J. (2004). *Quantitative finance and risk management*. World Scientific Publishing Co. Inc., River Edge, NJ. (Cited on page 580).
- [60] Dassios, A. and Lim, J. (2019). A variation of the Azéma martingale and drawdown options. *Mathematical Finance*, 29:1116–1130. (Cited on page 476).
- [61] Deelstra, G., Diallo, I., and Vanmaele, M. (2010). Moment matching approximation of Asian basket option prices. *J. Comput. Appl. Math.*, 234:1006–1016. (Cited on page 497).
- [62] Deelstra, G., Liinev, J., and Vanmaele, M. (2004). Pricing of arithmetic basket options by conditioning. *Insurance Math. Econom.*, 34:55–57. (Cited on page 497).
- [63] Demeterfi, K., Derman, E., Kamal, M., and Zou, J. (1999). More than you ever wanted to know about volatility swaps. Quantitative Strategies Research Notes. (Cited on page 359).
- [64] Derman, E. and Kani, I. (1994). Riding on a smile. *Risk Magazine*, 7(2):139–145. (Cited on page 374).
- [65] Devore, J. L. (2003). *Probability and Statistics for Engineering and the Sciences*. Duxbury Press, sixth edition edition. (Cited on page 819).
- [66] Di Nunno, G., Øksendal, B., and Proske, F. (2009). *Malliavin Calculus for Lévy Processes with Applications to Finance*. Universitext. Springer-Verlag, Berlin. (Cited on pages 111, 293, and 805).
- [67] Doob, J. (1953). *Stochastic processes*. John Wiley & Sons Inc., New York. (Cited on page 516).
- [68] Doob, J. (1984). *Classical potential theory and its probabilistic counterpart*. Springer-Verlag, Berlin. (Cited on page 516).
- [69] Dothan, L. (1978). On the term structure of interest rates. *Jour. of Fin. Ec.*, 6:59–69. (Cited on pages 618 and 637).
- [70] Downes, A., Joshi, M., and Denson, N. (2008). *Quant Job Interview Questions and Answers*. CreateSpace Independent Publishing Platform, first edition. (Cited on page 604).
- [71] Dudley, R. (2002). *Real analysis and probability*, volume 74 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge. Revised reprint of the 1989 original. (Cited on page 165).
- [72] Dufresne, D. (2000). Laguerre series for Asian and other options. *Math. Finance*, 10(4):407–428. (Cited on page 490).
- [73] Dufresne, D. (2001). The integral of geometric Brownian motion. *Adv. in Appl. Probab.*, 33(1):223–241. (Cited on page 487).
- [74] Dupire, B. (1994). Pricing with a smile. *Risk Magazine*, 7(1):18–20. (Cited on pages 374, 376, and 386).

- [75] Dvoretzky, A., Erdős, P., and Kakutani, S. (1961). Nonincrease everywhere of the Brownian motion process. In *Proc. 4th Berkeley Sympos. Math. Statist. and Prob., Vol. II*, pages 103–116. Univ. California Press, Berkeley, Calif. (Cited on page 390).
- [76] Einstein, A. (1905). Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen. *Annalen der Physik*, 17:549–560. (Cited on page 2).
- [77] El Karoui, N., Frachot, A., and Geman, H. (1997). A note on the behavior of long zero coupon rates in a no arbitrage framework. *Review of Derivatives Research*, 1:351–369. (Cited on page 689).
- [78] El Khatib, Y. (2003). Contributions to the study of discontinuous markets via the Malliavin calculus. PhD thesis, Université de La Rochelle. (Cited on page 472).
- [79] El Khatib, Y. and Privault, N. (2003). Computations of replicating portfolios in complete markets driven by normal martingales. *Applicationes Mathematicae*, 30:147–172. (Cited on page 470).
- [80] Elliott, R. and Kopp, P. (2005). *Mathematics of Financial Markets*. Springer Finance. Springer-Verlag, Berlin, second edition. (Cited on pages 545 and 556).
- [81] Eriksson, J. and Persson, J. (2006). Pricing turbo warrants. Preprint. (Cited on pages 65 and 425).
- [82] Faff, R. and Gray, P. (2006). On the estimation and comparison of short-rate models using the generalised method of moments. *Journal of Banking and Finance*, 30:3131–3146. (Cited on page 1185).
- [83] Feller, W. (1951). Two singular diffusion problems. *Ann. of Math. (2)*, 54:173–182. (Cited on pages 330 and 614).
- [84] Fernholz, E. (2002). *Stochastic Portfolio Theory*, volume 48 of *Applications of Mathematics (New York)*. Springer-Verlag, New York. Stochastic Modelling and Applied Probability. (Cited on page 222).
- [85] Filipović, D. (1999). A note on the Nelson-Siegel family. *Math. Finance*, 9(4):349–359. (Cited on page 681).
- [86] Folland, G. B. (1999). *Real analysis*. Pure and Applied Mathematics (New York). John Wiley & Sons Inc., New York, second edition. (Cited on page 153).
- [87] Föllmer, H. and Leukert, P. (1999). Quantile hedging. *Finance and Stochastics*, 3:251–273. (Cited on page 317).
- [88] Föllmer, H. and Schied, A. (2004). *Stochastic finance*, volume 27 of *de Gruyter Studies in Mathematics*. Walter de Gruyter & Co., Berlin. (Cited on pages 30, 36, 73, 74, 111, and 123).
- [89] Fonseca, J. D. and Martini, C. (2016). The α -hypergeometric stochastic volatility model. *Stochastic Process. Appl.*, 126(5):1472–1502. (Cited on page 355).
- [90] Fouque, J., Papanicolaou, G., and Sircar, K. (2000). *Derivatives in financial markets with stochastic volatility*. Cambridge University Press, Cambridge. (Cited on page 351).

- [91] Fouque, J., Papanicolaou, G., Sircar, K., and Sølna, K. (2011). *Multi-scale Stochastic Volatility for Equity, Interest Rate Derivatives, and Credit Derivatives*. Cambridge University Press, Cambridge. (Cited on pages 330, 351, 353, and 354).
- [92] Friz, P. and Gatheral, J. (2005). Valuation of volatility derivatives as an inverse problem. *Quant. Finance*, 5(6):531–542. (Cited on pages 338, 341, 379, and 1061).
- [93] Fukasawa, M., Maire, B., and Wunsch, M. (2023a). Model-free hedging of impermanent loss in geometric mean market makers. Preprint arXiv:2303.11118. (Cited on page 971).
- [94] Fukasawa, M., Maire, B., and Wunsch, M. (2023b). Weighted variance swaps hedge against impermanent loss. *Quant. Finance*, 23(6):901–911. (Cited on pages 971 and 1049).
- [95] Garman, M. and Kohlhagen, S. (1983). Foreign currency option valuations. *J. International Money and Finance*, 2:231–237. (Cited on pages 590 and 591).
- [96] Gatheral, J. (2006). *The Volatility Surface: A Practitioner's Guide*. Wiley. (Cited on pages 338, 351, 356, and 382).
- [97] Geman, H., El Karoui, N., and Rochet, J.-C. (1995). Changes of numéraire, changes of probability measure and option pricing. *J. Appl. Probab.*, 32(2):443–458. (Cited on pages 576 and 598).
- [98] Geman, H. and Yor, M. (1993). Bessel processes, Asian options and perpetuities. *Math. Finance*, 3:349–375. (Cited on pages 486, 490, and 1120).
- [99] Gerber, H. and Shiu, E. (1994). Option pricing by Esscher transforms. *Transactions of Society of Actuaries*, 46:99–191. (Cited on page 807).
- [100] Gerber, H. and Shiu, E. (1996). Martingale approach to pricing perpetual American options on two stocks. *Math. Finance*, 6(3):303–322. (Cited on page 566).
- [101] Glasserman, P. (2004). *Monte Carlo methods in financial engineering*, volume 53 of *Applications of Mathematics (New York)*. Springer-Verlag, New York. Stochastic Modelling and Applied Probability. (Cited on page 809).
- [102] Goldman, M., Sosin, H., and Gatto, M. (1979). Path dependent options: Buy at the low, sell at the high. *The Journal of Finance*, 34(5):1111–1127. (Cited on pages 409, 414, 453, and 462).
- [103] Gradshteyn, I. S. and Ryzhik, I. M. (2007). *Table of integrals, series, and products*. Elsevier/Academic Press, Amsterdam, seventh edition. (Cited on pages 341 and 387).
- [104] Gram, J. (1883). Über die Entwicklung reeller Funktionen in Reihen mittelst der Methode der kleinsten Quadraten. *J. Reine Angew. Math*, 94:41–73. (Cited on page 790).
- [105] Greene, W. (2003). *Econometric analysis*. Prentice Hall, fifth edition. (Cited on page 620).
- [106] Guirrerri, S. (2015). *YieldCurve: Modelling and Estimation of the Yield Curve*. R package version 4.1. (Cited on page 663).

- [107] Hagan, P., Kumar, D., Lesniewski, A., and Woodward, D. (2002). Managing smile risk. *Wilmott Magazine*, pages 84–108. (Cited on pages 330, 387, and 1058).
- [108] Hagan, P. and Lesniewski, A. (2019). Bartlett’s Delta in the SABR model. *Wilmott Magazine*, pages 54–61.
- [109] Han, J., Gao, M., Zhang, Q., and Li, Y. (2013). Option prices under stochastic volatility. *Appl. Math. Lett.*, 26(1):1–4. (Cited on page 355).
- [110] Hardy, G. H., Littlewood, J. E., and Pólya, G. (1988). *Inequalities*. Cambridge Mathematical Library. Cambridge University Press, Cambridge. Reprint of the 1952 edition. (Cited on page 518).
- [111] Harrison, J. and Kreps, D. (1979). Martingales and arbitrage in multi-period securities markets. *Journal of Economic Theory*, 20:341–408. (Cited on pages 73 and 74).
- [112] Harrison, J. and Pliska, S. (1981). Martingales and stochastic integrals in the theory of continuous trading. *Stochastic Process. Appl.*, 11:215–260. (Cited on pages 207 and 210).
- [113] Haug, E. (2006). *The complete guide to option pricing formulas*. McGraw Hill, second edition. (Cited on page 1106).
- [114] Heath, D., Jarrow, R., and Morton, A. (1992). Bond pricing and the term structure of interest rates: a new methodology. *Econometrica*, 60:77–105. (Cited on pages 4 and 674).
- [115] Hefter, M. and Herzurm, A. (2017). Optimal strong approximation of the one-dimensional squared Bessel process. *Commun. Math. Sci.*, 15(8):2121–2141. (Cited on page 190).
- [116] Henry-Labordère, P. (2009). *Analysis, Geometry, and Modeling in Finance*. Chapman & Hall/CRC Financial Mathematics Series. CRC Press, Boca Raton, FL. (Cited on page 387).
- [117] Heston, S. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343. (Cited on pages 330, 332, 347, 348, and 356).
- [118] Heston, S., Loewenstein, M., and Willard, G. (2007). Options and bubbles. *The Review of Financial Studies*, 20(2):359–390. (Cited on page 315).
- [119] Heynen, R. and Kat, H. (1996). Pricing and hedging power options. *Asia-Pacific Financial Markets*, 3:253–261. (Cited on page 1175).
- [120] Hiriart-Urruty, J.-B. and Lemaréchal, C. (2001). *Fundamentals of convex analysis*. Grundlehren Text Editions. Springer-Verlag, Berlin. (Cited on page 33).
- [121] Hirsch, F. and Lacombe, G. (1999). *Elements of functional analysis*, volume 192 of *Graduate Texts in Mathematics*. Springer-Verlag, New York. (Cited on page 163).
- [122] Ho, S. and Lee, S. (1986). Term structure movements and pricing interest rate contingent claims. *Journal of Finance*, 41:1011–1029. (Cited on page 618).

- [123] HSB Ltd. (2021). Launch announcement and supplemental listing document for warrants over single equities. 28 October. (Cited on pages 83 and 874).
- [124] HSB Ltd. (2022). Derivative warrants relating to the existing issued ordinary shares (“Shares”) of Tencent Holdings Limited (“Company”) (the “Warrants”) - Matters arising from the declaration of special interim dividend by the Company. 19 January. (Cited on page 83).
- [125] Huang, B. (2017). Analysis of contract-for-difference with Vasicek process and cointegrated system in stock market and Nordic electricity market. MSc Thesis, University of Oslo. (Cited on page 644).
- [126] Hull, J. and White, A. (1987). The pricing of options on assets with stochastic volatilities. *Journal of Finance*, 42(2):281–300. (Cited on page 357).
- [127] Hull, J. and White, A. (1990). Pricing interest rate derivative securities. *The Review of Financial Studies*, 3:537–592. (Cited on pages 618 and 674).
- [128] Ikeda, N. and Watanabe, S. (1989). *Stochastic Differential Equations and Diffusion Processes*. North-Holland. (Cited on pages 170 and 276).
- [129] Itô, K. (1944). Stochastic integral. *Proc. Imp. Acad. Tokyo*, 20:519–524. (Cited on page 2).
- [130] Itô, K. (1951). On stochastic differential equations. *Mem. Amer. Math. Soc.*, No. 4:51. (Cited on page 3).
- [131] Jacka, S. (1991). Optimal stopping and the American put. *Mathematical Finance*, 1:1–14. (Cited on page 557).
- [132] Jacod, J. and Protter, P. (2000). *Probability essentials*. Springer-Verlag, Berlin. (Cited on page 819).
- [133] Jacquier, A. (2017). Advanced methods in derivatives pricing with application to volatility modelling. https://www.ma.imperial.ac.uk/~ajacque/IC_AMDP/IC_AMDP_Docs/AMDP.pdf. Accessed: 2022-04-08. (Cited on pages 317 and 1034).
- [134] Jaillet, P., Lamberton, D., and Lapeyre, B. (1990). Variational inequalities and the pricing of American options. *Acta Appl. Math.*, 21:263–289. (Cited on page 556).
- [135] Jamshidian, F. (1989). An exact bond option formula. *The Journal of Finance*, XLIV(1):205–209. (Cited on pages 699 and 720).
- [136] Jamshidian, F. (1996). Sorting out swaptions. *Risk Magazine*, 9(3):59–60. (Cited on page 598).
- [137] Jarrow, R., Protter, P., and Shimbo, K. (2007). Asset price bubbles in complete markets. In *Advances in mathematical finance*, Appl. Numer. Harmon. Anal., pages 97–121. Birkhäuser Boston, Boston, MA. (Cited on page 315).
- [138] Jeanblanc, M. and Privault, N. (2002). A complete market model with Poisson and Brownian components. In Dalang, R., Dozzi, M., and Russo, F., editors, *Seminar on Stochastic Analysis, Random Fields and Applications (Ascona, 1999)*, volume 52 of *Progress in Probability*, pages 189–204. Birkhäuser, Basel. (Cited on page 805).

- [139] Jensen, J. (1906). Sur les fonctions convexes et les inégalités entre les valeurs moyennes. *Acta Math.*, 30:175–193. (Cited on pages [131](#), [482](#), [518](#), [553](#), and [706](#)).
- [140] Kakushadze, Z. (2015). Path integral and asset pricing. *Quant. Finance*, 15(11):1759–1771. (Cited on page [643](#)).
- [141] Kallenberg, O. (2002). *Foundations of Modern Probability*. Probability and its Applications. Springer-Verlag, New York, second edition. (Cited on page [852](#)).
- [142] Kemna, A. and Vorst, A. (1990). A pricing method for options based on average asset values. *Journal of Banking and Finance*, 14:113–129. (Cited on page [483](#)).
- [143] Keynes, J. (1924). *A Tract on Monetary Reform*. MacMillan & Co., London. (Cited on page [610](#)).
- [144] Kim, Y.-J. (2002). Option pricing under stochastic interest rates: An empirical investigation. *Asia-Pacific Financial Markets*, 9:23–44. (Cited on page [701](#)).
- [145] Klebaner, F. (2005). *Introduction to stochastic calculus with applications*. Imperial College Press, London, second edition. (Cited on pages [314](#), [1023](#), and [1134](#)).
- [146] Kloeden, P. and Platen, E. (1999). *Numerical Solution of Stochastic Differential Equations*, volume 23 of *Applications of Mathematics (New York)*. Springer-Verlag, New York. Stochastic Modelling and Applied Probability. (Cited on page [187](#)).
- [147] Korn, R., Korn, E., and Kroisandt, G. (2010). *Monte Carlo methods and models in finance and insurance*. Chapman & Hall/CRC Financial Mathematics Series. CRC Press, Boca Raton, FL. (Cited on page [809](#)).
- [148] Lamberton, D. and Lapeyre, B. (1996). *Introduction to stochastic calculus applied to finance*. Chapman & Hall, London. (Cited on pages [111](#) and [500](#)).
- [149] Leung, T. and Sircar, K. (2015). Implied volatility of leveraged ETF options. *Applied Mathematical Finance*, 22(2):162–188. (Cited on page [320](#)).
- [150] Levy, E. (1992). Pricing European average rate currency options. *Journal of International Money and Finance*, 11:474–491. (Cited on page [491](#)).
- [151] Lindström, E. (2007). Estimating parameters in diffusion processes using an approximate maximum likelihood approach. *Annals of Operations Research*, 151:269–288. (Cited on page [1186](#)).
- [152] Lipton, A. (2001). *Mathematical methods for foreign exchange. A financial engineer's approach*. World Scientific Publishing Co., Inc., River Edge, NJ. (Cited on page [225](#)).
- [153] Longstaff, F. and Schwartz, E. (2001). Valuing American options by simulation: a simple least-squares approach. *Review of Financial Studies*, 14:113–147. (Cited on pages [558](#) and [560](#)).
- [154] Lyashenko, A. and Mercurio, F. (2020). Looking forward to backward-looking rates: A modeling framework for term rates replacing LIBOR.

- Available at SSRN: <https://ssrn.com/abstract=3330240>. (Cited on page 678).
- [155] Lyuu, Y.-D. (2021). Course on principles of financial computing, Part 6. Barrier options. Lecture notes, National Taiwan University. (Cited on page 423).
 - [156] Mamon, R. (2004). Three ways to solve for bond prices in the Vasicek model. *Journal of Applied Mathematics and Decision Sciences*, 8(1):1–14. (Cited on page 1204).
 - [157] Margrabe, W. (1978). The value of an option to exchange one asset for another. *The Journal of Finance*, XXXIII(1):177–186. (Cited on pages 596 and 600).
 - [158] Marsh, T. and Rosenfeld, E. (1983). Stochastic processes for interest rates and equilibrium bond prices. *The Journal of Finance*, 38(2):635–646. Papers and Proceedings Forty-First Annual Meeting American Finance Association New York, N.Y. (Cited on pages 617 and 646).
 - [159] Matsumoto, H. and Yor, M. (2005). Exponential functionals of Brownian motion. I. Probability laws at fixed time. *Probab. Surv.*, 2:312–347. (Cited on page 487).
 - [160] Mel'nikov, A. and Petrachenko, Y. (2005). On option pricing in binomial market with transaction costs. *Finance and Stochastics*, 9:141–149. (Cited on page 907).
 - [161] Mel'nikov, A., Volkov, S., and Nechaev, M. (2002). *Mathematics of financial obligations*, volume 212 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI. Translated from the 2001 Russian original by H. H. McFaden. (Cited on page 317).
 - [162] Mercurio, F. (2018). A simple multi curve model for pricing SOFR futures and other derivatives. Working paper, Bloomberg LP. (Cited on page 716).
 - [163] Merton, R. (1973). Theory of rational option pricing. *Bell Journal of Economics*, 4(1):141–183. (Cited on page 597).
 - [164] Merton, R. (1976). Option pricing when underlying stock returns are discontinuous. *J. of Financial Economics*, 3:125–144. (Cited on page 796).
 - [165] Mikosch, T. (1998). *Elementary stochastic calculus—with finance in view*, volume 6 of *Advanced Series on Statistical Science & Applied Probability*. World Scientific Publishing Co. Inc., River Edge, NJ. (Cited on page 949).
 - [166] Milevsky, M. (1998). A closed-form approximation for valuing basket options. *Journal of Derivatives*, 55:54–61. (Cited on page 497).
 - [167] Milne, J. (2005). Mathematical apocrypha. <https://www.jmilne.org/math/apocrypha.html>. Accessed: 2020-06-12. (Cited on page 1262).
 - [168] MOE and UCLES (2022a). Mathematics (Syllabus 4048), Singapore-Cambridge General Certificate of Education Ordinary Level. https://www.seab.gov.sg/docs/default-source/national-examinations/syllabus/olevel/2022syllabus/4048_y22_sy.pdf. Accessed: 2024-04-20. (Cited on page 827).

- [169] MOE and UCLES (2022b). Mathematics (Syllabus 9758), Singapore-Cambridge General Certificate of Education Advanced Level Higher 2. https://www.seab.gov.sg/docs/default-source/national-examinations/syllabus/alevel/2022syllabus/9758_y22_sy.pdf. Accessed: 2022-12-28. (Cited on page 827).
- [170] Mohan, V. (2022). Automated market makers and decentralized exchanges: a DeFi primer. *Financial Innovation*, 8:20. (Cited on page 229).
- [171] Mörters, P. and Peres, Y. (2010). *Brownian Motion*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, Cambridge. (Cited on page 391).
- [172] Nelson, C. and Siegel, A. (1987). Parsimonious modeling of yield curves. *Journal of Business*, 60:473–489. (Cited on pages 655 and 679).
- [173] Neuberger, A. (1994). The log contract. *Journal of Portfolio Management*, 20(2):74–80. (Cited on page 359).
- [174] Norris, J. (1998). *Markov Chains*, volume 2 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press, Cambridge. Reprint of 1997 original. (Cited on page 741).
- [175] Øksendal, B. and Sulem, A. (2005). *Applied stochastic control of jump diffusions*. Springer-Verlag, Berlin. (Cited on page 760).
- [176] Papanicolaou, A. and Sircar, K. (2014). A regime-switching Heston model for VIX and S&P 500 implied volatilities. *Quant. Finance*, 14(10):1811–1827. (Cited on pages 330 and 379).
- [177] Peng, S. (2010). Backward stochastic differential equation, nonlinear expectation and their applications. In *Proceedings of the International Congress of Mathematicians. Volume I*, pages 393–432. Hindustan Book Agency, New Delhi. (Cited on page 311).
- [178] Pintoux, C. and Privault, N. (2010). A direct solution to the Fokker-Planck equation for exponential Brownian functionals. *Analysis and Applications*, 8(3):287–304. (Cited on page 639).
- [179] Pintoux, C. and Privault, N. (2011). The Dothan pricing model revisited. *Math. Finance*, 21:355–363. (Cited on page 638).
- [180] Pitman, J. (1999). *Probability*. Springer. (Cited on page 819).
- [181] Prayoga, A. and Privault, N. (2017). Pricing CIR yield options by conditional moment matching. *Asia-Pacific Financial Markets*, 24:19–38. (Cited on pages 333 and 641).
- [182] Privault, N. (2008). Stochastic analysis of Bernoulli processes. *Probab. Surv.*, 5:435–483. arXiv:0809.3168v3. (Cited on page 111).
- [183] Privault, N. (2009). *Stochastic analysis in discrete and continuous settings: with normal martingales*, volume 1982 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin. (Cited on pages 111, 113, 167, 170, 275, and 293).
- [184] Privault, N. (2018). *Understanding Markov Chains*. Springer Undergraduate Mathematics Series. Springer, second edition. (Cited on page 742).

- [185] Privault, N. (2021a). Notes on Financial Risk and Analytics. Course notes, https://personal.ntu.edu.sg/nprivault/MH8331/financial_risk_analytics.pdf, 292 pages. Accessed: 2021-09-08. (Cited on page 169).
- [186] Privault, N. (2021b). *Stochastic Interest Rate Modeling With Fixed Income Derivative Pricing (3rd edition)*. Advanced Series on Statistical Science & Applied Probability. World Scientific Publishing Co., Singapore. 372 pp. (Cited on pages 674, 682, 684, 686, 701, 716, and 730).
- [187] Privault, N. and She, Q. (2016). Option pricing and implied volatilities in a 2-hypergeometric stochastic volatility model. *Appl. Math. Lett.*, 53:77–84. (Cited on page 355).
- [188] Privault, N. and Teng, T.-R. (2012). Risk-neutral hedging of interest rate derivatives. *Risk and Decision Analysis*, 3:201–209. (Cited on pages 598, 704, and 715).
- [189] Privault, N. and Uy, W. (2013). Monte Carlo computation of the Laplace transform of exponential Brownian functionals. *Methodol. Comput. Appl. Probab.*, 15(3):511–524. (Cited on page 639).
- [190] Privault, N. and Wei, X. (2009). Calibration of the LIBOR market model - implementation in PREMIA. Bankers, Markets & Investors, 99:20–28. (Cited on page 716).
- [191] Privault, N. and Yu, J. (2016). Stratified approximations for the pricing of options on average. *Journal of Computational Finance*, 19(4):95–113. (Cited on pages 494 and 640).
- [192] Profeta, C., Roynette, B., and Yor, M. (2010). *Option prices as probabilities*. Springer Finance. Springer-Verlag, Berlin. A new look at generalized Black–Scholes formulae. (Cited on page 404).
- [193] Protter, P. (2001). A partial introduction to financial asset pricing theory. *Stochastic Process. Appl.*, 91(2):169–203. (Cited on pages 295 and 599).
- [194] Protter, P. (2004). *Stochastic integration and differential equations*, volume 21 of *Stochastic Modelling and Applied Probability*. Springer-Verlag, Berlin, second edition. (Cited on pages 180, 186, 282, 287, 288, 295, 582, 595, 628, 629, and 1029).
- [195] Rebonato, R. (2009). *The SABR/LIBOR Market Model Pricing, Calibration and Hedging for Complex Interest-Rate Derivatives*. John Wiley & Sons. (Cited on page 331).
- [196] Revuz, D. and Yor, M. (1994). *Continuous Martingales and Brownian Motion*. Springer-Verlag. (Cited on pages 153, 189, 193, 929, and 1030).
- [197] Rogers, L. and Shi, Z. (1995). The value of an Asian option. *J. Appl. Probab.*, 32(4):1077–1088. (Cited on page 502).
- [198] Ross, S. (2015). The recovery theorem. *The Journal of Finance*, 70:615–648. (Cited on pages 48 and 84).
- [199] Rouah, F. (2013). *The Heston Model and its Extensions in Matlab and C#*. Wiley Finance. John Wiley & Sons, Inc. (Cited on page 348).
- [200] Rubinstein, M. (1991). Pay now, choose later. *Risk Magazine*, 4:13–13. (Cited on page 304).

- [201] Rubinstein, M. and Reiner, E. (1991). Breaking down the barriers. *Risk*, 4(9):28–35. (Cited on pages 435 and 438).
- [202] Rudin, W. (1974). *Real and Complex Analysis*. McGraw-Hill. (Cited on pages 163 and 165).
- [203] Ruiz de Chávez, J. (2001). Predictable representation of the binomial process and application to options in finance. In *XXXIII National Congress of the Mexican Mathematical Society (Spanish) (Saltillo, 2000)*, volume 29 of *Aportaciones Mat. Comun.*, pages 223–230. Soc. Mat. Mexicana, México. (Cited on page 111).
- [204] Samuelson, P. (1965). Rational theory of warrant pricing. *Industrial Management Review*, 6(2):13–39. (Cited on page 3).
- [205] Santa-Clara, P. and Sornette, D. (2001). The dynamics of the forward interest rate curve with stochastic string shocks. *The Review of Financial Studies*, 14(1):149–185. (Cited on page 694).
- [206] Sato, K. (1999). *Lévy processes and infinitely divisible distributions*, volume 68 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge. (Cited on page 775).
- [207] Schoenmakers, J. (2005). *Robust LIBOR modelling and pricing of derivative products*. Chapman & Hall/CRC Financial Mathematics Series. Chapman & Hall/CRC, Boca Raton, FL. (Cited on pages 715 and 716).
- [208] Schroeder, T. and Coffey, B. (2018). Assessment of physical delivery mechanisms on the live cattle futures market. Research Report. (Cited on page 9).
- [209] Scorsese, M. (2013). *The Wolf of Wall Street*. Paramount Pictures. (Cited on page 364).
- [210] Sharpe, W. (1978). *Investments*. Prentice Hall, Englewood Cliffs, N.J. (Cited on page 75).
- [211] Shiryaev, A. (1999). *Essentials of stochastic finance*. World Scientific Publishing Co. Inc., River Edge, NJ. (Cited on pages 207 and 210).
- [212] Shreve, S. (2004). *Stochastic calculus for finance. II*. Springer Finance. Springer-Verlag, New York. Continuous-time models. (Cited on pages 404, 413, 432, 506, 542, 562, 607, and 1083).
- [213] Steele, J. (2001). *Stochastic Calculus and Financial Applications*, volume 45 of *Applications of Mathematics*. Springer-Verlag, New York. (Cited on page 554).
- [214] Stroock, D. (2011). *Probability theory, an analytic view*. Cambridge University Press, Cambridge, second edition. (Cited on page 851).
- [215] Stähli, P. and Maringer, D. (2025). Start-to-low drawdown as a risk measure and its application to portfolio optimization for levered investors under solvency regimes. Working Paper 2025/07, University of Basel, Center of Business and Economics (WWZ), Basel. (Cited on page 1068).
- [216] Svensson, L. (1994). Estimating and interpreting forward interest rates: Sweden 1992–1994. National Bureau of Economic Research Working Paper 4871. (Cited on pages 655 and 679).

- [217] Tanaka, K., Yamada, T., and Watanabe, T. (2010). Applications of Gram-Charlier expansion and bond moments for pricing of interest rates and credit risk. *Quant. Finance*, 10(6):645–662. (Cited on page 790).
- [218] Tencent Ltd (2021). Declaration of interim dividend by way of distribution in specie of class A ordinary shares of JD.com, Inc. 23 December. (Cited on page 83).
- [219] Thiele, T. (1899). On semi invariants in the theory of observations (Om Iagttagelseslærens Halvinvarianter). *Kjöbenhavn Overs.*, pages 135–141. (Cited on page 789).
- [220] Turnbull, S. and Wakeman, L. (1992). A quick algorithm for pricing European average options. *Journal of Financial and Quantitative Analysis*, 26:377–389. (Cited on page 491).
- [221] Üstünel, A. (2009). Probabilistic solution of the American options. *J. Funct. Anal.*, 256:3091–3105. (Cited on page 556).
- [222] Vašíček, O. (1977). An equilibrium characterisation of the term structure. *Journal of Financial Economics*, 5:177–188. (Cited on pages 4, 194, 609, 610, 630, 661, 699, 933, 1186, and 1209).
- [223] Večeř, J. (2001). A new PDE approach for pricing arithmetic average Asian options. *Journal of Computational Finance*, 4:105–113. (Cited on page 506).
- [224] Watson, G. N. (1995). *A treatise on the theory of Bessel functions*. Cambridge University Press, Cambridge. Reprint of the second (1944) edition. (Cited on page 640).
- [225] Widder, D. (1975). *The heat equation*. Academic Press, New York. Pure and Applied Mathematics, Vol. 67. (Cited on page 256).
- [226] Wiener, N. (1923). Differential space. *Journal of Mathematics and Physics of the Massachusetts Institute of Technology*, 2:131–174. (Cited on page 2).
- [227] Williams, D. (1991). *Probability with martingales*. Cambridge Mathematical Textbooks. Cambridge University Press, Cambridge. (Cited on page 111).
- [228] Wilmott, P. (2006). *Paul Wilmott on Quantitative Finance*. John Wiley & Sons. (Cited on page 565).
- [229] Wong, H. and Chan, C. (2008). Turbo warrants under stochastic volatility. *Quant. Finance*, 8(7):739–751. (Cited on pages 65 and 425).
- [230] Wu, X. (2000). A new stochastic duration based on the Vasicek and CIR term structure theories. *Journal of Business Finance and Accounting*, 27. (Cited on page 653).
- [231] Yang, Z., Ewald, C.-O., and Menkens, O. (2011). Pricing and hedging of Asian options: quasi-explicit solutions via Malliavin calculus. *Math. Methods Oper. Res.*, 74:93–120. (Cited on page 511).
- [232] Yor, M. (1992). On some exponential functionals of Brownian motion. *Adv. in Appl. Probab.*, 24(3):509–531. (Cited on pages 329, 487, and 638).
- [233] Zhang, P. (1995). Correlation digital options. *Journal of Financial Engineering*, 4(1):75–96. (Cited on page 1106).

Index

σ -algebra, 820

σ -field, 820

 code, 11, 155–157, 159, 163, 169, 171, 187–189, 213, 217, 220, 232, 233, 239, 241, 247, 248, 255, 264, 265, 267, 268, 281, 290, 344, 365–367, 373, 377, 382, 383, 392, 425, 427, 432, 434, 436, 494, 529, 588, 611–613, 615, 616, 622, 623, 634, 636, 663, 664, 682, 717, 736, 739, 742, 743, 746, 750, 761, 762, 772, 784–787, 791, 810, 811, 837, 840, 1031, 1063, 1064, 1080, 1087, 1144

 package

bizdays, 255

fOptions, 366, 559, 1080, 1144

quantmod, 232, 362, 427, 494, 529, 621, 663, 762, 783

RQuantLib, 717

Sim.DiffProc, 233, 623

YieldCurve, 663

absence of arbitrage, 27, 624

abstract Bayes formula, 577

accreting swap, 670

adapted process, 168

adjusted close price, 232, 763

adjustment

convexity, 358

admissible portfolio strategy, 205

affine model, 618

affine PDE, 347, 618, 1192

American

binary option

finite expiration, 568

perpetual, 568

forward contract, 570

option

butterfly, 561

call, 537, 547

dividend, 562, 565

finite expiration, 552

perpetual, 537, 547

put, 537

amortizing swap, 670

annuity

measure, 708, 728, 730

numéraire, 708, 726

annuity numéraire, 669

approximation

gamma, 640

lognormal, 490

arbitrage

absence, 27, 624

calendar, 385

continuous time, 205

discrete time, 62

forex, 23

maturity, 385

opportunity, 24

price, 15, 42, 74, 90, 286

strike, 385

triangular, 23

arcsine distribution, 418

arithmetic average, 479

Asian

forward contract, 304, 509

Asian option, 479, 481

basket, 497

call, 479

dividends, 512

hedging, 511

asset pricing

first theorem, 29

- continuous time, 207, 674
- discrete time, 73
- second theorem, 36
 - continuous time, 210
 - discrete time, 74
- at the money, 63, 98, 253, 305
- ATM, 63, 98, 253, 305
- attainable, 34, 40, 89, 210
- automated market maker (AMM), 51, 226
- Bachelier model, 192, 216, 263, 271, 299, 306, 388, 692, 1056
- backward
 - induction, 100, 102
 - stochastic differential equation, 311
- backward-looking bond price, 667
- Bank for International Settlements, 8
- Barone–Adesi & Whaley approximation, 559, 561, 1144
- barrier
 - level, 424
- barrier forward contract, 449
 - down-and-in
 - long, 449, 1091
 - down-and-out
 - long, 449, 1092
 - up-and-in
 - long, 449, 1087
 - up-and-out
 - long, 449, 1089
- barrier option, 65, 421, 424
 - down-and-in
 - call, 425, 438
 - put, 425, 440
 - down-and-out
 - call, 425, 434, 447
 - put, 425, 436
 - down-knock-in forward, 451
 - hedging, 446
 - in-out parity, 425
 - up-and-in
 - call, 425, 439
 - put, 425, 441
 - up-and-out
 - call, 425, 426
 - put, 425, 432
 - up-knock-in forward, 451
- basis point, 719
- basket option, 300, 497
- BDT model, 651
- bear spread option, 301, 992
- Bermudan swaption, 717
- Bernoulli
 - distribution, 834
- Bessel function, 614, 640
- BGM model, 687, 702
- binary
 - tree, 75, 135
- binary option, 64, 129, 270, 308, 606, 891
 - American
 - finite expiration, 568
 - perpetual, 568
 - barrier, 450
- binomial
 - distribution, 834
 - model, 75
 - dividends, 138
 - transaction costs, 135
- BIS, 8
- bisection method, 268, 365
- bizdays ( package), 255
- Black
 - (1976) formula, 704
 - LIBOR caplet formula, 702
 - SOFR caplet formula, 707
 - SOFR swaption formula, 715
 - swaption formula, 713
- Black–Derman–Toy model, 651
- Black–Scholes
 - calibration, 369
 - formula, 261, 267, 289, 592, 688
 - call options, 237, 703, 708
 - put options, 245, 246, 704
 - PDE, 235, 260, 268, 444, 447, 815
 - with jumps, 799
- bond
 - convertible, 647, 865
 - convexity, 652
 - corporate, 647
 - duration, 648, 652
 - immunization, 1195
 - ladder, 708
 - option, 605, 699, 1176
 - pricing
 - PDE, 628, 685
 - yield, 637
 - zero-coupon, 623
- Borel–Cantelli Lemma, 199, 824
- boundary condition, 815
- box spread option, 302
- break-even
 - rate, 669, 670
 - strike price, 52
 - underlying asset price, 99, 253, 991
- Brent, 11
- Bretton Woods, 574
- bridge model, 690
- Brownian

- bridge, 194, 643
- extrema, 408
- motion, 149
 - geometric, 274
 - Lévy's construction, 157, 196, 964
 - series construction, 154, 158
- BSDE, 311
- bubble, 315
- bull spread option, 301, 992
- business time, 255
- butterfly option, 303, 996
 - American, 561
- buy back guarantee, 8, 594
- buy limit, 521
- calendar arbitrage, 385
- calendar time, 255
- call
 - level, 424
 - option, 8
 - price, 425, 432
 - spread collar option, 131
 - swaption, 713
- call-put parity, 128, 247, 291, 310, 378, 592, 1099
- callable
 - bear contract, 65, 424–426
 - bull contract, 425, 432
- Cantor function, 392
- Cantor set, 392
- cap pricing, 705
- Capital Asset Pricing Model (CAPM), 313, 1022
- caplet pricing, 701
- cash settlement, 7, 9, 62, 245
- cash-or-nothing option, 64, 606
- Category 'N' CBBC, 425
- Category 'R' CBBC, 425, 448
- cattle futures, 237
- Cauchy
 - distribution, 831
 - sequence, 944
- CBBC, 65, 424–426, 432
 - Category 'N', 425
 - Category 'R', 425, 448
 - rebate, 425, 448
 - residual, 425, 448
- CBOE, 613
 - Volatility Index[®], 378
- CEV model, 317, 617
- CFD, 644
- change of measure, 282
- change of numéraire, 576, 592
- characteristic
 - function, 846
- Charles relation, 176
- Chi square distribution, 307, 614, 1009
- Chicago Board Options Exchange, 613
- chooser option, 310, 1014
- CIR model, 195, 264, 307, 614, 647
- CKLS model, 644
- Clark–Ocone formula, 113, 470
- cliquet
 - option, 305
- closing portfolio value, 58
- collar option, 11
 - call spread, 131
 - costless, 13
 - put spread, 130
- complement rule, 823
- complete market, 36, 40, 285
- complete space, 165, 174
- completeness
 - continuous time, 209
 - discrete time, 73
- compound Poisson
 - martingale, 775
 - process, 744, 779, 792
- compounded yield to maturity, 652
- compounding
 - linear, 666, 667
- conditional
 - expectation, 66, 839, 848
 - probability, 824, 825
- conditioning, 824
- constant product automated market maker (CPAMM), 51, 226
- constant repayment, 54
- contingent claim, 33, 62, 73, 89
 - attainable, 34, 40, 210
- continuous-time
 - limit, 123
- contract for difference, 644
- conversion rate, 647
- convertible bond, 647, 865
- convexity, 652
- convexity adjustment, 358
- corporate bond, 647
- correction
 - convexity, 358
- correlation
 - perfect, 684, 694
 - problem, 683
- correlation option, 451
- cost of carry, 588
- costless collar option, 13
- counterparty risk, 102
- counting process, 735, 737

- coupon
 - bond, 624
 - rate, 635
- Courtadon model, 617, 646
- Cox process, 739
- Cox–Ingersoll–Ross model, 195
- Cox–Ross–Rubinstein model, 75, 239
 - dividends, 138
 - transaction costs, 135
- credit exposure, 102
- critical price, 562
- CRR model, 75, 239
 - dividends, 138
 - transaction costs, 135
- cumulants, 764
 - Gaussian, 790
 - Poisson, 790
- cumulative distribution function, 830
 - joint, 397, 833
- cup & handle, 1
- date
 - of payment, 265
 - of record, 265
- decentralized exchange (DEX), 51, 226
- deflated price, 575
- Delta, 103, 106, 235, 240, 242, 248, 250, 267, 271, 297, 996, 1045
 - hedging, 272, 295, 599, 600
 - Vega hedging, 357
- Delta-Vega hedging, 357
- density
 - function, 829
 - marginal, 398, 833
- derivatives
 - fixed income, 695
 - interest rate, 695
 - market, 8
- differential
 - inequalities, 544
 - interest rate, 312
- diffusion
 - elasticity, 617, 646
- digital option, 64, 129, 270, 308, 606, 891
- discounting, 61, 203
 - lemma, 214, 285, 793
- discrete distribution, 834
- dispersion index, 641, 738
- distribution
 - arcsine, 418
 - Bernoulli, 834
 - binomial, 834
 - Cauchy, 831
 - discrete, 834
 - exponential, 831
 - gamma, 831
 - Gaussian, 830
 - geometric, 835
 - Hartman–Watson, 488
 - invariant, 345, 354, 611, 615
 - leptokurtic, 787
 - lognormal, 122, 220, 490, 639, 831, 988
 - marginal, 842
 - negative binomial, 835
 - Pascal, 835
 - Poisson, 835
 - stable, 788
 - stationary, 345, 354, 611, 615
 - uniform, 830
- dividend, 83, 138, 141, 264, 304, 512, 562, 565
 - date of payment, 265
 - date of record, 265
 - ex-date, 265
 - payable date, 265
- dollar value, 652
- dominated convergence theorem, 540, 549, 1032
- Doob–Meyer decomposition, 534
- Dothan model, 618, 637
- down-knock-in forward option, 451
- drawdown
 - option, 476
 - process, 453, 462
- drift estimation, 361
- drifted Brownian motion, 279
- Dupire PDE, 374
- duration, 648, 652
- early exercise premium, 545
- ECB, 664
- effective gearing, 106, 253
- efficient market hypothesis, 1, 72, 995
- elasticity, 254
- elasticity of diffusion, 617, 646
- entitlement ratio, 10, 243, 249, 369–372
- entropy
 - contract, 359
 - swap, 384
- equivalent probability measure, 29, 37, 73, 207, 284
- Esscher transform, 807
- ETF, 313, 381, 1064
- Euclidean path integral, 642
- Euler discretization, 809
- EURIBOR, 665
- European option
 - knock-in, 451

- knock-out, 451
- event, 820
- ex-dividend, 265, 512
- excess kurtosis, 790
- exchange option, 566, 595
- exchange-traded fund, 313, 381, 1064
- exercise price, 6
- exotic option, 65, 95, 293, 421
 - Asian, 479
 - continuous time, 421, 453, 479
 - discrete time, 111
 - lookback option, 453
- expectation, 836
 - conditional, 839, 848
- exponential
 - distribution, 741, 831
 - Lévy model, 795
 - Vasicek model, 194, 617, 933
- extrinsic value, 98, 252
- face value, 623, 652
- Fano factor, 641
- Fatou's lemma, 276, 525, 846, 1022
- FED, 665
- Feller condition, 614
- filtration, 67, 151, 515
- finite differences
 - explicit scheme, 812, 815
 - implicit scheme, 814, 816
- first theorem of asset pricing, 29, 73, 207, 674
- fixed
 - income, 609
 - derivatives, 695
 - leg, 669
 - rate, 701, 706
- floating
 - leg, 669
 - rate, 701, 706
 - strike, 65
- floorlet, 704, 720
- folded normal distribution, 395
- fOptions (R package), 366, 559, 1080, 1144
- foreign exchange, 586
 - option, 590
- foreign exchange option, 225
- forex arbitrage, 23
- formula
 - Lévy-Khintchine, 747
 - smoothing, 752
 - Tanaka, 196, 226, 941, 963
 - Taylor, 986
- forward
 - contract, 132, 236, 268, 289, 604, 655, 689, 978, 1170
 - American, 570, 1162
 - Asian, 304, 509
 - non-deliverable, 237
 - measure, 648, 696
 - price, 575
 - range, 128, 268
 - rate, 655, 689
 - agreement, 655, 689
 - spot, 655, 656, 658, 701, 706
 - swap, 668
 - start option, 305
 - swap rate, 668
- forward measure, 576
- forward swap
 - measure, 708, 728, 730
- four-way collar option, 11
- Fourier
 - synthesis, 158
 - transform, 347
 - inversion, 347
- FRA, 655, 689
- Fubini theorem, 754
- fugazi (the), 364
- futures contract, 237, 644, 897
- FX option, 225
- gains process, 94
- Galton board, 121
- Gamma
 - Greek, 242, 250, 1045
 - process, 761
- gamma
 - approximation, 640
 - distribution, 831
 - function, 831
 - swap, 359
- Gamma hedging, 272
- gap, 783
- Garman-Kohlagen formula, 590
- Gaussian
 - cumulative distribution function, 125, 700
 - distribution, 238, 830
 - random variable, 847
- gearing, 98, 253
 - effective, 106, 253
- Geman-Yor method, 490
- generating function, 195, 847
- geometric
 - average, 482, 510
 - Brownian motion, 216, 274
 - distribution, 835

- geometric mean market maker model, 51, 226
- Girsanov Theorem, 282, 313, 582
 - jump processes, 769, 793
- Goldman–Sosin–Gatto formula, 409, 414, 453, 462
- Greeks, 250, 1045
 - Delta, 235, 240, 242, 248, 250, 267, 271, 297, 996, 1045
 - Gamma, 242, 250, 1045
 - Rho, 250, 1045
 - Rhod, 271
 - Theta, 250, 309, 1014, 1045
 - Vanna, 1044
 - Vega, 250, 271, 450, 1045, 1095
- gross market value, 8
- gross world product, 8
- guarantee
 - buy back, 8, 594
 - price lock, 10
- GWP, 8
- Hartman–Watson distribution, 488
- Hawaiian option, 480
- heat
 - equation, 255, 812
 - map, 400
- hedge and forget, 236, 896, 1171, 1256
- hedge ratio, 107, 254
- hedging, 35, 101, 102, 111, 292
 - change of numéraire, 597
 - Delta, 272, 357
 - Gamma, 272
 - gamma, 272
 - mean-variance, 801
 - quantile, 317
 - static, 236, 896, 1171, 1256
 - strategy, 294
 - Vega, 357
 - Velta, 357
 - with jumps, 801
- Heston model, 330, 356
- hexanomial model, 920
- HIBOR, 665
- historical
 - probability measure, 280
 - volatility, 331, 361
- hitting
 - time, 521
- HJM
 - condition, 674
 - model, 672
 - SOFR model, 678
- Ho–Lee model, 618
- Hull–White model, 618, 674
- immunization, 1195
- implied
 - probability, 17
 - volatility, 364
- in the money, 63, 98, 253, 371, 885
- in-out parity, 425, 1099
- inclusion-exclusion principle, 823
- independence, 824, 826, 829, 833, 835, 840, 847, 852
- independent increments, 274, 772
- indicator function, 827
- infimum, 835
- infinitesimal, 177
- information flow, 68
- instantaneous forward rate, 658
- interest rate
 - derivatives, 695
 - differential, 312
 - model
 - affine, 618
 - Constant Elasticity of Variance, 617
 - Courtadon, 617, 646
 - Cox–Ingersoll–Ross, 264, 307, 614
 - Dothan, 618, 637
 - exponential Vasicek, 194, 617, 933
 - Ho–Lee, 618
 - Hull–White, 618, 674
 - Marsh–Rosenfeld, 617, 646
 - Vasicek, 610
 - short term, 609
- intrinsic value, 43, 98, 252
- invariant distribution, 345, 354, 611, 615
- inverse Gaussian process, 762
- IPython notebook, 15, 75, 96, 100, 103, 108, 110, 137, 145, 157, 158, 239, 268, 365, 369, 392, 508, 561, 613, 615, 680, 907, 919
- Itô
 - formula, 179, 307
 - pathwise, 756
 - with jumps, 757
- isometry, 161, 164, 172, 753
- process, 179, 181, 234, 981
- stochastic integral, 160, 170, 172, 273
- table, 182, 457
 - with jumps, 758, 1244
- ITM, 63, 98, 253, 371, 885
- Jamshidian’s trick, 720
- Jensen’s inequality, 132, 276, 483, 518, 896, 990, 1124
- joint

- cumulative distribution function, 397, 833
- probability density function, 832
- jump-diffusion process, 783
- kimchi, 23
- knock-in option, 451
- knock-out option, 65, 425, 451
- Kullback-Leibler entropy, 794
- kurtosis, 787, 790
- Lévy
 - construction of Brownian motion, 157, 196, 964
 - process, 761
- Lévy characterization, 193
- Lévy-Khintchine formula, 747
- Lagrangian, 641
- Laplace transform, 490, 534
- law
 - of total expectation, 842
 - of total probability, 823, 826, 842
- least square Monte Carlo, 558
- least square regression, 619
- leg
 - fixed, 669
 - floating, 669
- Leibniz integral rule, 675
- lemma
 - Neyman–Pearson, 318
- leptokurtic distribution, 787
- leverage, 224, 314, 320, 1022, 1039
- liability, 14
- LIBOR
 - model, 665
 - rate, 665
 - swap rate, 669, 711, 713
- LIBOR-SOFR swap, 716
- Lipschitz function, 595
- liquidity
 - pool, 51, 226
 - provider, 51, 226
- local
 - time, 196
 - volatility, 372, 815
- local martingale, 315
- log
 - contract, 269, 305, 359
 - option, 321
 - return, 362
 - dynamics, 217, 358, 795
 - variance, 122, 220, 328
- log variance, 220
- lognormal
 - approximation, 490
 - distribution, 122, 220, 639, 831, 988
 - long box spread option, 302, 994
 - long forward contract, 805, 806
 - lookback option
 - call, 462
 - put, 453, 456
 - LOTUS, 837
 - LSM, 558
 - Macauley duration, 652
 - Mandatory Call Event, 448
 - marginal
 - density, 398, 833
 - distribution, 842
 - Margrabe formula, 596
 - mark to market, 42, 74, 90, 237, 286, 897
 - market
 - bubble, 315
 - completeness, 36, 40, 73
 - efficiency, 995
 - making, 42
 - price of risk, 278, 284, 351, 628
 - volatility, 351
 - market terms and data, 98, 250
 - Markov property, 186, 595, 599
 - Marsh–Rosenfeld model, 617, 646
 - martingale, 66, 152, 273, 516
 - compound Poisson, 775
 - continuous time, 151, 206
 - discrete time, 69
 - local, 315
 - measure
 - continuous time, 205, 792
 - discrete time, 72
 - method, 285
 - Poisson, 771, 772
 - submartingale, 516
 - supermartingale, 516
 - transform, 70, 94, 275, 522
 - maturity, 6
 - transformation, 656
 - maturity arbitrage, 385
 - maximum of Brownian motion, 389
 - MCE, 448
 - mean
 - hitting time, 531
 - reversion, 609
 - mean-square distance, 851
 - measurability, 168
 - Merton model, 796
 - method
 - bisection, 365
 - Newton–Raphson, 365

- Milstein discretization, 810
- Minkowski inequality, 165
- model
 - affine, 618
 - Bachelier, 216, 263, 271, 299, 306, 388, 1056
 - Barone-Adesi & Whaley, 561
 - BDT, 651
 - binomial, 75
 - dividends, 138
 - transaction costs, 135
 - CEV, 317, 617
 - CIR, 614, 647
 - CKLS, 644
 - Courtadon, 617, 646
 - Dothan, 618, 637
 - exponential Lévy, 795
 - exponential Vasicek, 617
 - hexanominal, 920
 - Ho–Lee, 618
 - Hull–White, 618, 674
 - Marsh–Rosenfeld, 617, 646
 - Merton, 796
 - pentanominal, 920
 - trinomial, 82, 126, 143
 - Vasicek, 610
- modified
 - Bessel function, 614, 640
 - duration, 652
- moment
 - generating function, 847, 1248
- moment matching, 490
- moments
 - Gaussian, 790
 - Poisson, 790
- moneyness, 63, 98, 253
- moving average, 480
- MPoR, 278, 284, 351, 628
- Musiela notation, 673
- natural logarithm, 238
- negative
 - binomial distribution, 835
 - inverse Gaussian process, 762
 - premium, 29
 - risk premium, 206
- Nelson–Siegel, 679, 682
- Newton–Raphson method, 365
- Neyman–Pearson Lemma, 318
- nominal value, 652
- non-deliverable forward contract, 237
- noncentral Chi square, 307, 614, 1009
- nonlocal operator, 801
- notional, 670
 - principal, 719, 1219, 1220
- notional amount, 8
- numéraire, 206, 573
 - annuity, 669, 708
 - invariance, 597
- numéraire invariance, 599
- OLS, 619
- opening
 - jump, 783
 - portfolio price, 58
- optimal stopping, 552
- option
 - Asian, 479
 - basket, 497
 - call, 479
 - at the money, 305
 - barrier, 65, 421
 - basket, 300, 497
 - bear spread, 301, 992
 - binary, 64, 129, 606, 891
 - box spread, 302
 - bull spread, 301, 992
 - butterfly, 303, 996
 - cash-or-nothing, 64, 606
 - chooser, 310, 1014
 - cliquet, 305
 - digital, 64, 129, 606, 891
 - drawdown, 476
 - effective gearing, 106, 253
 - exotic, 65, 95, 111, 293, 421, 453, 479
 - extrinsic value, 98, 252
 - foreign exchange, 225
 - forward start, 305
 - gearing, 98, 253
 - Hawaiian, 480
 - intrinsic value, 98, 252
 - issuer, 15, 35
 - knock-in, 451
 - knock-out, 65, 425
 - long box spread, 302, 994
 - lookback, 453
 - on average, 64, 299, 479, 510
 - on extrema, 422
 - out of the money, 309
 - path-dependent, 111, 293
 - power, 132, 216, 266, 306, 604, 897
 - premium, 35, 99, 253
 - ratchet, 305
 - straddle, 1018
 - tunnel, 126, 128
 - vanilla, 235
 - variance call, 336
 - variance swap, 332

- volatility swap, 341, 387, 1061
- writer, 15, 35
- zero-collar, 13
- optional
 - sampling, 522
 - stopping, 522
- order book, 966
- Ornstein–Uhlenbeck process, 610
- OTM, 63, 98, 253, 309
- out of the money, 63, 98, 253, 309
- Paley–Wiener series, 158
- par value, 623, 652
- parity
 - call-put, 128, 247, 291, 310, 378, 592, 1099
 - in-out, 425, 1099
- Partial integro-differential equation, 799
- partition, 826, 849
- Pascal distribution, 835
- path
 - freezing, 721
 - integral, 98, 422, 580, 641
 - Euclidean, 642
- path-dependent option, 111, 293
- pathwise Itô formula, 756
- payable date, 265
- payer
 - swap, 669
 - swaption, 710
- payoff function, 7, 9, 421
- PDE
 - affine, 347, 618, 1192
 - Black–Scholes, 235, 260
 - Heston, 345
 - integro-differential, 799
 - variational, 556
- pentanomial model, 920
- perfect correlation, 684, 694
- Perron–Frobenius theorem, 84
- physical delivery, 7, 9, 62, 245
- PIDE, 799
- Planck constant, 641
- Poisson
 - compound martingale, 744, 792
 - distribution, 835
 - process, 735
 - compound, 779
- portfolio, 22
 - process, 94
 - replicating, 103, 108
 - strategy, 34, 56, 89, 207, 210
 - admissible, 205, 210
 - update, 208, 211
- value, 60, 90
- power option, 132, 216, 266, 306, 604, 897
- predictable process, 70, 93, 752
- premium
 - early exercise, 545
 - kimchi, 23
 - negative, 29
 - option, 99, 253
 - risk, 29, 206, 278
- price
 - critical, 562
 - graph, 7, 9, 12, 130, 131, 893, 894
- price lock guarantee, 10
- pricing, 89, 95
 - with jumps, 793
- principal amount, 719
- probability
 - conditional, 824, 825
 - density function, 829
 - joint, 832
 - distribution, 828
 - measure, 823
 - equivalent, 29, 37, 73, 207, 284
 - sample space, 819
 - space, 824
- process
 - counting, 735
 - Cox, 739
 - drawdown, 453, 462
 - gamma, 761
 - inverse Gaussian, 761
 - Lévy, 761
 - predictable, 70, 93, 752
 - stable, 761
 - stopped, 521
 - variance gamma, 761
- pushforward measure, 775
- put
 - option, 5
 - spread collar option, 130
 - swaption, 715
- Python code, 15, 75, 96, 100, 103, 108, 110, 137, 145, 146, 157, 158, 239, 268, 365, 369, 392, 508, 561, 613, 615, 680, 736, 739, 742–744, 746, 751, 837, 840, 907, 919
- Python package
 - yfinance, 369
- quantile hedging, 317
- Quantlib, 717
- quantmod ( package), 232, 362, 427, 494, 529, 621, 663, 762, 783

- Radon–Nikodym, 280
- Radon–Nikodym density, 576
- random
 - product, 844
 - sum, 843
 - variable, 827
- range forward, 128, 268
- ratchet
 - option, 305
- rate
 - forward, 655, 689
 - forward swap, 668
 - instantaneous forward, 658
 - LIBOR, 665, 670
 - swap, 669
 - LIBOR swap, 711, 713
 - SOFR swap, 715
 - swap, 668
- realized variance, 331, 362
 - swap, 332
- receiver swaption, 715
- recovery theorem, 48, 84
- reflection principle, 421
- relative entropy, 794
- renewal processes, 743
- replicating portfolio, 103, 108
- replication, 35
- repo market, 667
- repurchase agreement, 667
- return
 - log, 362
- Rho, 250, 271, 1045
- Riccati equation, 633, 1197
- risk
 - counterparty, 102
 - market price, 278, 284, 351, 628
 - premium, 29, 206
- risk premium, 278
- risk-neutral
 - measure, 16, 28, 792
 - continuous time, 205, 278
 - discrete time, 72
 - probabilities, 16
- riskless asset, 125, 213
- Robinhood incident, 995
- RQuantLib, 717
- running maximum, 389, 390
- SABR model, 356
- second theorem of asset pricing, 36, 74, 210
- self-financing portfolio
 - change of numéraire, 599
 - continuous time, 207, 209, 210, 212, 801
 - discrete time, 58, 90
 - sell stop, 521
 - seller swap, 669
 - share right, 28
 - Sharpe ratio, 284
 - SHIBOR, 665
 - short rate, 609
 - short selling, 40, 107, 244
 - ratio, 23
 - SIBOR, 665
 - Sim.DiffProc, 233, 623
 - singular measure, 457
 - skewness, 787, 789
 - slow-fast system, 352
 - smile, 366
 - smooth pasting, 541
 - smoothing formula, 752
 - SOFR, 667
 - Black caplet formula, 707
 - caplet, 706
 - forward rate, 668
 - HJM model, 678
 - swap rate, 672, 715
 - swaption, 715
 - spline function, 377
 - spot forward rate, 655, 656, 658, 701, 706
 - spread option, 692
 - square-integrable
 - functions, 160
 - random variables, 164
 - St. Petersburg paradox, 838
 - stability warrant, 450
 - stable
 - distribution, 788
 - process, 762
 - static hedging, 236, 896, 1171, 1256
 - stationary distribution, 345, 354, 611, 615
 - stochastic
 - calculus, 176
 - differential equations, 186
 - integral, 92, 158, 168
 - with jumps, 751
 - integral decomposition, 113, 192, 292, 295
 - process, 56
 - stochastic portfolio theory, 222
 - stop-loss/start-gain strategy, 225
 - stopped process, 521, 522
 - stopping time, 520
 - theorem, 522
 - straddle option, 1018
 - Stratonovich integral, 949
 - strike arbitrage, 385
 - strike price, 6, 34

- floating, 65
- string model, 694
- strong Markov property, 741
- submartingale, 516
- super-hedging, 35, 74
- supermartingale, 516
- Svensson parametrization, 679
- swap, 668
 - amortizing, 670
 - entropy, 384
 - forward measure, 708, 728, 730
 - gamma, 359
 - payer, 669
 - rate, 668, 670
 - seller, 669
 - variance, 332
- swaption, 710
 - Bermudan, 717
- Tanaka formula, 196, 226, 941, 963
- Taylor's formula, 177, 986
- telescoping sum, 671
- tenor structure, 575, 668, 695
- terms and data, 98, 250
- ternary tree, 82, 126, 143
- theorem
 - asset pricing, 29, 36, 73, 74, 207, 210, 674
 - dominated convergence, 540, 549
 - Fubini, 754
 - Girsanov, 282, 313, 582, 769, 793
 - Perron–Frobenius, 84
 - recovery, 48, 84
 - stopping time, 522
- Theta, 250, 309, 1014, 1045
- TIBOR, 665
- time
 - business, 255
 - splitting, 225, 307, 962
- tower property, 69, 71, 93, 100, 173, 274, 276, 295, 298, 577, 627, 842, 846, 852, 876, 919, 1204
- transaction cost, 135
- transform
 - Esscher, 807
 - Fourier, 347
 - Laplace, 490, 534
 - martingale, 70, 94, 522
- treasury note, 613
- tree
 - binary, 75, 135
 - ternary, 82, 126, 143
- trend estimation, 361
- triangle inequality, 165
- triangular arbitrage, 23
- trinomial model, 82, 126, 143
- tunnel option, 126, 128
- turbo warrant, 65, 424–426, 432
- two-asset correlation option, 451
- two-factor model, 685
- uniform distribution, 830
- Uniswap, 51, 226
- up-knock-in forward option, 451
- valuation period, 448
- vanilla option, 63, 95, 235
- vanna, 1044
- variable rate, 701, 706
- variance, 843
 - call option, 336
 - gamma process, 761
 - realized, 331, 362
 - swap, 332
- variational PDE, 544, 556
- Vasicek model, 610
- Vega, 250, 271, 450, 1045, 1095
 - notional, 332
- VIX[®], 378
- volatility
 - historical, 331, 361
 - implied, 364
 - level, 332, 384
 - local, 372, 815
 - smile, 366
 - surface, 369
 - swap, 341, 387, 1061
 - variance swap, 332
- warrant, 10, 83, 243
 - stability, 450
 - terms and data, 255
- turbo, 65, 424–426, 432
- weak solution, 189
- West Texas Intermediate (WTI), 5, 11
- Wiener space, 2
- yfinance (Python package), 369
- yield, 655, 658, 701, 706
 - bond, 637
 - compounded to maturity, 652
 - curve, 656
 - data, 663
 - inversion, 665
- YieldCurve (R package), 663
- zero measure, 392
- zero-
 - collar option, 13
 - coupon bond, 623

Author index

- Achdou, Y. 377
 Ackerer, D. 377
 Albanese, C. 330, 614
 Albrecher, H. 348
 Allegretto, J. 559
 Applebaum, D. 761
 Aristotle 5
 Armstrong, J. 272
 Aste, T. 229
 Attari, M. 348

 Bachelier, L. 2, 158
 Barone-Adesi, G. 559, 561
 Barrieu, P. 488
 Bartlett, B. 357, 1044
 Benth, F.E. 497
 Bergomi, L. 1051
 Bermin, H. 469, 470
 Bickerseth, M. 668
 Billingsley, R.S. 995
 Björk, T. 44, 682
 Black, F. 3, 210, 231, 651, 702, 707, 713
 Borodin, A.N. 401
 Bosq, D. 737
 Boulding, K.E. 213, 574
 Boyle, P.P. 907
 Brace, A. 4, 687
 Breeden, D.T. 373
 Brémaud, P. 752
 Brigo, D. 341, 630, 686, 1193
 Briola, A. 229
 Broadie, M. 358, 565
 Brody, D.C. 645
 Brown, C. 511
 Brown, R. 2, 152
 Buchen, P. 1106
 Burdzy, K. 390
 Buterin, V. 51, 226

 Carmona, R. 307, 692
 Carr, P. 334, 359, 489, 490
 Chan, C.M. 65, 425
 Chan, K.C. 644
 Chance, D.M. 995
 Charlier, C.V.L. 790

 Charpentier, A. 663
 Chataigner, M. 377
 Christensen, B.J. 682
 Christensen, J.H.E. 682
 Çınlar, E. 829
 Coffey, B.K. 9
 Cont, R. 761, 768, 774, 783, 805
 Courtadon, G. 617, 646
 Cousin, A. 377
 Cox, A.M.G. 315
 Cox, J.C. 75, 264, 307, 614
 Cramér, H. 790
 Crépey, S. 480
 Curran, M. 495

 Da Fonseca, J. 355
 Dahl, L. O. 497
 Dana, R.A. 462
 Dash, J. 580
 Dassios, A. 476
 Deelstra, G. 497
 Demeterfi, K. 359
 Denson, N. 604
 Derman, E. 359, 374, 651
 Detemple, J. 565
 Devore, J.L. 819
 Di Nunno, G. 111, 293, 805
 Diallo, I. 497
 Diebold, F.X. 682
 Ding, Y. 668
 Dixon, M. 377
 Doob, J.L. 516, 522, 534
 Dothan, L.U. 618, 637
 Downes, A. 604
 Dudley, R.M. 165
 Dufresne, D. 487, 490
 Dupire, B. 374
 Durrleman, V. 307, 692
 Dvoretzky, A. 390

 Einstein, A. 2
 El Karoui, N. 690
 El Khatib, Y. 470, 472
 Elliott, R.J. 545, 556, 559
 Erdos, P. 390
 Eriksson, J. 65, 425

- Ewald, C.-O. 511
- Faff, R. 1185
- Feller, W. 330, 614, 1009
- Fernholz, E.R. 222, 955
- Filipović, D. 682
- Folland, G.B. 153
- Föllmer, H. 111, 317
- Fouque, J.-P. 330, 351
- Frachot, A. 690
- Friz, P. 338, 379
- Fukasawa, M. 971, 1049
- Galton, F. 121
- Gao, M. 355
- Garman, M.B. 590
- Gatarek, D. 4, 687
- Gatheral, J. 338, 351, 356, 379, 382
- Gatto, M.A. 409, 414, 453, 462
- Geman, H. 486, 490, 576, 598, 690, 1120
- Gerber, H.U. 566, 807
- Glasserman, P. 809
- Goldman, M.B. 409, 414, 453, 462
- Gradshteyn, I.S. 341, 1061
- Gram, J.P. 790
- Gray, P. 1185
- Greene, W.H. 620
- Gueye, D. 377
- Guirrerri, S. 663
- Hagan, P.S. 330, 357, 387, 1058
- Han, J. 355
- Handley, J.C. 511
- Hardy, G.H. 518
- Harrison, J.M. 73, 74, 207, 210
- Haug, E.G. 1106
- Heath, D. 4, 674
- Heftner, M. 190
- Henry-Labordère, P. 387
- Herzwurm, A. 190
- Heston, S.L. 315, 330, 347, 348, 357
- Heynen, R.C. 1175
- Hiriart-Urruty, J.-B. 33
- Hirsch, F. 163
- Ho, S.Y. 618
- Hobson, D.G. 315
- Huang, B. 644
- Hughston, L.P. 645
- Hull, J. 357, 618
- Ikeda, N. 170, 276
- Ingersoll, J.E. 264, 307, 614
- Itô, K. 3
- Jacka, S.D. 557
- Jacod, J. 819
- Jacquier, A. 317, 1034
- Jaillet, P. 556
- Jain, A. 358
- Jamshidian, F. 598, 699, 720
- Jarrow, R. 4, 674
- Jarrow, R.A. 315
- Jeanblanc, M. 462, 805
- Jensen, J. 131, 482, 518, 553, 706
- Joshi, M.S. 604
- Kakushadze, Z. 643
- Kakutani, S. 390
- Kallenberg, O. 852
- Kamal, M. 359
- Kani, I. 374
- Karolyi, G.A. 644
- Kat, H.M. 1175
- Kemna, A.G.Z. 483
- Keynes, J.M. 610
- Kim, Y.-J. 701
- Klebaner, F. 314, 1023, 1134
- Kloeden, P.E. 187
- Kohlhagen, S.W. 590
- Kopp, P.E. 545, 556
- Korn, E. 809
- Korn, R. 809
- Kreps, D.M. 73, 74
- Kroisandt, G. 809
- Kumar, D. 330, 387, 1058
- Lacombe, G. 163
- Lamberton, D. 111, 500, 556
- Lapeyre, B. 111, 500, 556
- Lawi, S. 330, 614
- Lee, R. 334, 359
- Lee, S.B. 618
- Lemaréchal, C. 33
- Lesniewski, A. 357
- Lesniewski, A.S. 330, 387, 1058
- Leukert, P. 317
- Leung, T. 320
- Levy, E. 491
- Lévy, P. 193
- Li, Y. 355
- Liinev, J. 497
- Lim, J.W. 476
- Lindström, E. 1186
- Ling, C.-T. 511
- Lipton, A. 225
- Littlewood, J.E. 518
- Litzenberger, R.h. 373
- Loewenstein, M. 315

- Longstaff, F.A. 558, 560, 644
 Lyashenko, A. 678
 Lyuu, Y.-D. 423
- Maire, B. 971, 1049
 Mamon, R.S. 1204
 Margrabe, W. 596
 Maringer, D.G. 1068
 Marsh, T.A. 617, 646
 Martini, C. 355
 Matsumoto, H. 487
 Mayer, P. A. 348
 Meier, D.M. 645
 Melnikov, A.V. 317, 907
 Menkens, O. 511
 Mercurio, F. 341, 630, 686, 716, 1193
 Merton, R.C. 4, 597, 796
 Meyer, P.A. 534
 Mikosch, T. 949
 Milevsky, M.A. 497
 Milne, J.S. 1262
 Mörters, P. 391
 Morton, A. 4, 674
 Musiela, M. 4, 673, 687
- Nechaev, M.L. 317
 Nelson, C.R. 679
 Neuberger, A. 359
 Nguyen, H.T. 737
 Nikodym, O.M. 280
 Norris, J.R. 741
 Novikov, A. 282
- Øksendal, B. 111, 293, 761, 805
- Paley, R. 158
 Palmer, K.J. 511
 Papanicolaou, A. 330, 379
 Papanicolaou, G. 330, 351
 Peng, S. 311
 Peres, Y. 391
 Persson, J. 65, 425
 Petrachenko, Y.G. 907
 Pintoux, C. 638, 639
 Pironneau, O. 377
 Pitman, J. 819
 Platen, E. 187
 Pliska, S.R. 207, 210
 Poisson, S.D. 735
 Pólya, G. 518
 Prayoga, A. 333, 641
 Profeta, C. 404
 Proske, F. 111, 293, 805
- Protter, P. 180, 186, 282, 295, 315, 582, 595, 599, 628, 629, 819, 1029
- Radon, J. 280
 Rebonato, R. 331
 Reiner, E. 435, 438
 Revuz, D. 153, 1030
 Rochet, J.-C. 576, 598
 Rogers, C. 502
 Rosenfeld, E.R. 617, 646
 Ross, S. 48, 84
 Ross, S.A. 75, 264, 307, 614
 Rouah, F.D. 348
 Rouault, A. 488
 Roynette, B. 404
 Rubinstein, M. 75, 435, 438
 Rudebusch, G.D. 682
 Rudin, W. 163
 Ruiz de Chávez, J. 111
 Rutkowski, M. 668
 Ryzhik, I.M. 341, 1061
- Samuelson, P.A. 3
 Sanders, A.B. 644
 Santa-Clara, P. 694
 Sato, K. 775
 Schied, A. 30, 36, 73, 74, 111, 123
 Schoenmakers, J. 715, 716
 Scholes, M. 3, 4, 210, 231
 Schoutens, W. 348
 Schröder, M. 489, 490
 Schroeder, T.C. 9
 Schwartz, E.S. 558, 560
 Scorsese, M. 364
 Sharpe, W.F. 75
 She, Q.H. 355
 Shi, Z. 502
 Shimbo, K. 315
 Shiryayev, A.N. 207, 210
 Shiu, E.S.W. 566, 807
 Shreve, S. 404, 413, 432, 506, 542, 562, 607, 1083
 Siegel, A.F. 679
 Sircar, K.R. 320, 330, 351, 379
 Sølna, K. 330, 351
 Sornette, D. 694
 Sosin, H.B. 409, 414, 453, 462
 Stähli 1068
 Steele, J.M. 554
 Stroock, D.W. 851
 Sulem, A. 761
 Svensson, L.E.O. 679
- Tagasovska, N. 377

- Tanaka, K. 790
 Tankov, P. 761, 768, 774, 783, 805
 Tatlow, G. 272
 Teng, T.-R. 598, 704, 715
 Thales 5
 Thiele, T.N. 789
 Tistaert, J. 348
 Toy, B. 651
 Turnbull, S.M. 491

 UCLES, MOE & 827
 Üstünel, A.S. 556
 Uy, W.I. 639

 Vanmaele, M. 497
 Vašíček, O. 4, 610, 630
 Vatter, T. 377
 Večeř, J. 506
 Vidal-Tomás, D. 229
 Volkov, S.N. 317
 Vorst, A.C.F. 483
 Vorst, T. 907

 Wakeman, L. 491
 Wang, Y. 229

 Watanabe, S. 170, 276
 Watanabe, T. 790
 Wei, X. 716
 Whaley, R.E. 559, 561
 White, A. 357, 618
 Widder, D.V. 256
 Wiener, N. 2, 158
 Willard, G.A. 315
 Williams, D. 111
 Wilmott, P. 565
 Wong, H.Y. 65, 425
 Woodward, D.E. 330, 387, 1058
 Wu, X. 653
 Wunsch, M. 971, 1049

 Yamada, T. 790
 Yang, Z. 511
 Yor, M. 153, 330, 404, 486–488, 490, 638, 1030, 1120
 Yu, J.D. 494, 640

 Zhang, P.G. 1106
 Zhang, Q. 355
 Zou, J. 359

This text is an introduction to the pricing and hedging of financial derivatives, including vanilla and exotic options, by stochastic calculus and partial differential equation methods. The presentation is done both in discrete and continuous-time financial models, with an emphasis on the complementarity between algebraic and probabilistic methods. In particular it covers the pricing of some interest rate derivatives, of American options, of exotic options such as barrier, lookback and Asian options, and stochastic models with compound Poisson jumps. The text is accompanied with a number of figures and simulations, and includes numerous examples based on actual market data. The concepts presented are also illustrated by 384 figures, and 290 exercises and 18 problems with complete solutions. It also includes 45 Python codes and 104  scripts for illustrations based on market data.